MATH6183 – SQL Coursework

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Contents

1 Introduction **1**

2 SQL Solutions **2**

3 References **10**

1 Introduction

This report indicates an understanding of data retrieval, aggregation, and reporting using SQL. The aims are to comprehend the supplied Hospital Staff Shift Allocation database and answer SQL questions. The answer for each question is presented in section two, and there are explanations in the queries' comments and the explanation subsection.

In addition to subqueries, several queries in this report use Common Table Expressions (CTEs) to improve readability and maintainability. CTEs allow the creation of temporary virtual tables during query execution, making complex logic easier to understand and manage. Figure 1 below shows the Entity Relationship Diagram (ERD) of the database.

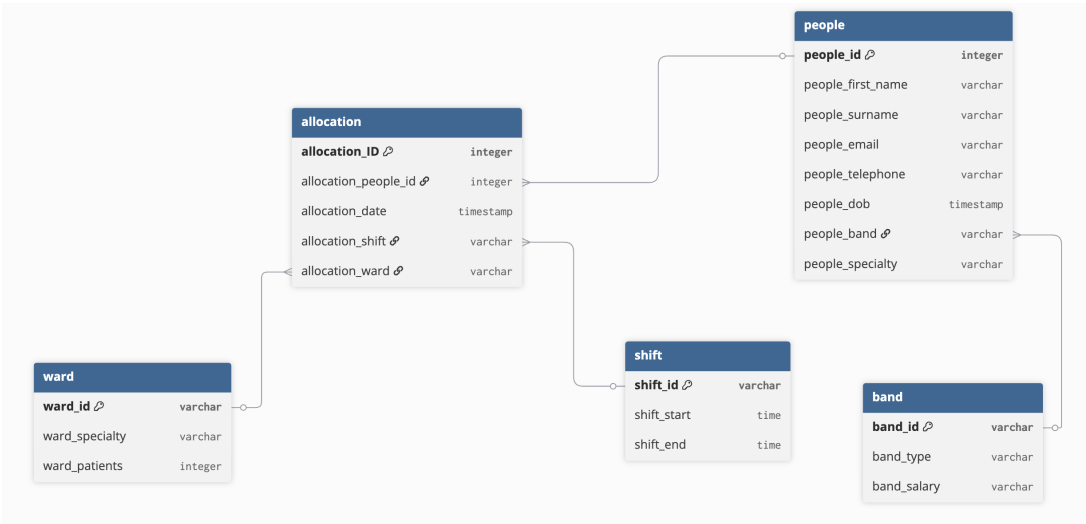

Figure 1: The Entity Relationship Diagram (ERD) of Hospital Staff Shift Allocation Database

2 SQL Solutions

Question 1

Task: Output the total number of hours that the staff member with ID number 10798 has worked in the given time period. Each shift lasts for 8 hours.

SQL Query: Total hours worked by staff member 10798

```sql
WITH shift_allocation AS (
  SELECT DISTINCT-- remove duplicate values in the same shift with different wards
    allocation_people_id,
    allocation_date,
    allocation_shift
  FROM allocation
  WHERE allocation_people_id ='10798' -- filter staff ID
)

SELECT
  COUNT(allocation_shift)*8 AS working_hours -- number of shifts * 8 hours
FROM shift_allocation
```

Explanation: First, the CTE "shift_allocation" extracts staff ID, date, and shift allocated to staff 10798, and the DISTINCT function removes duplication for cases where this member has a shift allocated to different wards. After that, the main query calculates total working hours by counting total shifts and multiplying by 8 hours.

Results: 2800 hours in total

working_hours
2800

Question 2

Task: Output a list of staff members aged over 65 on 1 January 2024 in alphabetical order of surname, listing their full names written as "[first name] [surname]" (e.g. John Smith) and their dates of birth

SQL Query: The list of staff members who have exceeded 65 years old

```sql
SELECT
  TRIM(people_first_name) || ' ' || TRIM(people_surname) AS full_name, -- get full name of staff member
  people_dob -- date of birth
FROM people
WHERE DATE(people_dob, '+65 years') <= '2024-01-01' -- get staff aged 65 and above
ORDER BY people_surname -- alphabetical order of surname
```

Explanation: The query retrieves the date of birth and full name data by combining first name and surname in the people table. The WHERE clause is to filter members older than 65. Finally, the results are sorted A-Z by surname.

Results: 30 rows & 2 columns

full_name	people_dob
ISAAC BELL	1958-03-04
TYLER CHRISTIE	1957-12-21
TEDDY CRAWFORD	1958-09-29
HOLLY DICKSON	1958-03-10
JACK DONALDSON	1958-04-07
NOAH FLEMING	1957-11-16
MARTHA GRANT	1957-09-23
KIAN HAMILTON	1958-12-24
ROBERT HAMILTON	1958-07-07
WILLIAM HENDERSON	1958-09-08
KATIE HUGHES	1958-08-17
AARON HUNTER	1957-10-17
GEORGE JOHNSTON	1958-06-13
LUKE KELLY	1958-07-12
MEGAN MACKAY	1958-06-26
RILEY MACKENZIE	1958-01-14
OLIVIA MACLEOD	1957-12-01
JULIA MCKAY	1958-09-13
MIA MILLAR	1958-06-17
ARTHUR MURRAY	1957-12-19
BETHANY REID	1958-05-03
EMILY ROBERTSON	1958-09-24
ETHAN ROBERTSON	1958-10-19
AIDEN SMITH	1958-08-28
RYAN SMITH	1957-09-22
GEORGIA STEVENSON	1957-09-15
ISABELLA STEWART	1958-07-28
MILLIE WALLACE	1957-10-07
BLAKE WILSON	1958-07-14
MILLIE WOOD	1958-11-06

Question 3

Task: Output a list of staff who were working in either of the two cardiology wards on 1 March 2024 giving their Staff ID, full names written as "[first name] [surname]" (e.g. John Smith) and the ward on which they were working. Where a staff member worked on the same ward multiple times, only include their name and ward once. Where they worked on multiple wards, return an entry for each ward that they worked on. Order by Ward ID and then surname.

SQL Query:

```sql
SELECT DISTINCT -- remove duplications of wards in different shifts / dates
  allocation.allocation_people_id AS staff_id,
  TRIM(people.people_first_name) || ' ' || TRIM(people.people_surname) AS full_name,
  allocation.allocation_ward AS ward_id
FROM allocation
INNER JOIN people
  ON people.people_id = allocation.allocation_people_id
WHERE allocation.allocation_ward IN ('C1', 'C2') -- filter cardiology wards
AND allocation.allocation_date = '2024-03-01' -- filter date
ORDER BY allocation.allocation_ward, people.people_surname -- sort by ward ID and surname
```

Explanation: The query returns a list of staff IDs, full Names, and ward IDs from the allocation table, joining them with the people table. The WHERE clause filters people in the cardiology wards on 03 Jan 2024. The results are then sorted by ward ID and surname A-Z

Results: 32 rows & 3 columns

staff_id	full_name	ward_id
10099	FLORENCE ALLAN	C1
10964	EVAN BELL	C1
10019	JAMIE DUNCAN	C1
10680	GABRIEL DUNCAN	C1
10856	LAYLA FORBES	C1
10308	CHARLIE GRANT	C1
10186	LILY HENDERSON	C1
10793	WILLIAM HENDERSON	C1
10166	ARTHUR JAMIESON	C1
10959	RILEY MACKENZIE	C1
10242	KAI MACLEOD	C1
10962	CONNOR MARSHALL	C1
10296	CAMERON MARTIN	C1
10049	MATILDA MILNE	C1
10616	ANNA RITCHIE	C1
10302	LOLA RUSSELL	C1
10694	JULIA SIMPSON	C1
10076	LEAH SUTHERLAND	C1
10838	ERIN WHITE	C1
10964	EVAN BELL	C2
10793	WILLIAM HENDERSON	C2
10186	LILY HENDERSON	C2
10636	WILLIAM KERR	C2
10547	DARCEY MACKAY	C2
10959	RILEY MACKENZIE	C2
10162	MICHAEL MORRISON	C2
10640	MADISON SCOTT	C2
10694	JULIA SIMPSON	C2
10240	HOLLIE WALLACE	C2
10172	GRACE WILLIAMS	C2
10908	VICTORIA WOOD	C2
10895	SARAH WRIGHT	C2

Question 4

Task: The hospital spotted suspicious behaviour on the Geriatric Wards (GC1 and GC2) on the following shifts: morning 23 January 2024; morning 17 April 2024; morning 21 June 2024; morning 18 August 2024. Return the full name(s) and Staff ID numbers of any staff members who have worked on all four of these shifts in either GC1 or GC2. Do not return the names of staff members who have only worked on one or two of these shifts.

SQL Query: Identify staff member who has worked in four suspicious shifts in Geriatric Wards

```sql
SELECT
  TRIM(people.people_first_name) || ' ' || TRIM(people.people_surname) as full_name,
  allocation.allocation_people_id AS staff_id
FROM allocation
INNER JOIN people
  ON people.people_id = allocation.allocation_people_id
WHERE allocation.allocation_shift IN ('Morning') -- focus on morning shifts only
    -- filter suspicious shift dates
  AND allocation.allocation_date IN ('2024-01-23', '2024-04-17', '2024-06-21', '2024-08-18')
  AND allocation.allocation_ward IN ('GC1', 'GC2') -- filter Geriatric wards
GROUP BY full_name, staff_id
-- filter only members who work on all these four dates
HAVING count(DISTINCT allocation.allocation_date) = 4
```

Explanation: The query displays members' full names and IDs, filtering by morning shift, specific dates, and Geriatric wards to find people with unusual behaviours. The HAVING condition guarantees that only staff members who worked across all four specified dates are returned in the results.

Results:

full_name	staff_id
JACK KENNEDY	10134

Question 5

Task: Output the number of each staff type (consultant, doctor, health care assistant, nurse) working each shift in either of the orthopaedics wards (OR1 or OR2) on 10 April 2024.

SQL Query:

```sql
SELECT
  band.band_type,
  allocation.allocation_shift,
    -- count members of staff for each type and shift
  COUNT(DISTINCT allocation.allocation_people_id) AS num_staff
FROM allocation
INNER JOIN people
  ON people.people_id = allocation.allocation_people_id
INNER JOIN band
  ON band.band_id = people.people_band
WHERE allocation.allocation_ward IN ('OR1', 'OR2') -- filter orthopaedics wards
  AND allocation.allocation_date = '2024-04-10' -- filter date: 10 April 2024
GROUP BY allocation.allocation_shift, band.band_type
ORDER BY band.band_type, allocation.allocation_shift
```

Explanation: The query extracts the number of staff by staff type (band type) by allocation shift. The WHERE conditions are to ensure the wards are orthopaedics and the date is 10 April 2024. The results are sorted A-Z by allocation shift and band type to make it organized

Results:

band_type	allocation_shift	num_staff
Consultant	Evening	2
Consultant	Morning	2
Consultant	Night	3
Doctor	Evening	2
Doctor	Morning	2
Doctor	Night	2
Health Care Assistant	Evening	6
Health Care Assistant	Morning	3
Health Care Assistant	Night	4
Nurse	Evening	5
Nurse	Morning	4
Nurse	Night	4

Question 6

Task: Output the total number of hours worked by each type of staff member (health care assistant, nurse, doctor, consultant) for each month from January to August 2024.

SQL Query: Total working hours by month by band type

```sql
WITH cleanup_table AS (

  SELECT DISTINCT -- remove duplications of wards in different shifts and dates
    allocation.allocation_date,
    allocation.allocation_people_id,
    allocation.allocation_shift,
    band.band_type
  FROM allocation
  INNER JOIN people -- acts as a connection between allocation and band
    ON people.people_id = allocation.allocation_people_id
  INNER JOIN band
    ON band.band_id = people.people_band
  WHERE allocation.allocation_date BETWEEN '2024-01-01' AND '2024-08-31' -- filter Jan-Aug 2024
)

SELECT
    -- create month format: mm-yyyy
  strftime('%m', allocation_date) || '-' || strftime('%Y', allocation_date) as month_num,
  band_type,
  count(allocation_shift)*8 as total_working_hours -- get total working hours by month
FROM cleanup_table
GROUP BY month_num, band_type
ORDER BY month_num
```

Explanation: The query returns total working hours by month by band type. The CTE "cleanup_table" ensures no duplication regarding different wards allocated in the same shifts and dates, and the WHERE condition filters the month between January and August 2024. Next, the main query calculates total working hours by counting total shifts multiplied by 8 hours, grouped by month and band type. In addition, the date is reformatted to mm-yyyy and ordered by month to make the result cleaner.

Results: 32 rows, 3 columns

month_num	band_type	total_working_hours
01-2024	Consultant	15160
01-2024	Doctor	14944
01-2024	Health Care Assistant	44528
01-2024	Nurse	36504
02-2024	Consultant	14096
02-2024	Doctor	13976
02-2024	Health Care Assistant	41896
02-2024	Nurse	33944
03-2024	Consultant	14800
03-2024	Doctor	14872
03-2024	Health Care Assistant	44976
03-2024	Nurse	36392
04-2024	Consultant	14576
04-2024	Doctor	14400
04-2024	Health Care Assistant	43512
04-2024	Nurse	35520
05-2024	Consultant	14816
05-2024	Doctor	14816
05-2024	Health Care Assistant	44880
05-2024	Nurse	36616
06-2024	Consultant	14344
06-2024	Doctor	14376
06-2024	Health Care Assistant	43808
06-2024	Nurse	35376
07-2024	Consultant	15000
07-2024	Doctor	14848
07-2024	Health Care Assistant	45024
07-2024	Nurse	36320
08-2024	Consultant	15040
08-2024	Doctor	14824
08-2024	Health Care Assistant	44840
08-2024	Nurse	36472

Question 7

Task: Output the average staff costs per shift for each specialty between 1 January 2024 and 31 August 2024. Report average costs to two decimal places. The average staff costs of a specialty are equal to the number of shifts worked by each staff member for that specialty multiplied by their cost per shift divided by the total number of shifts worked for that specialty by any member of staff.

SQL Query: The average staff costs per shift for each specialty

```sql
WITH shift_per_staff AS (
  SELECT
    allocation.allocation_people_id,
    allocation.allocation_date,
    allocation.allocation_shift,
    people.people_band,
    people.people_specialty,
        -- calculate cost per shift and rounded to 2 decimal places
    round(band.band_salary/230.0,2) AS cost_per_shift,
    count(allocation.allocation_ward) AS num_wards_allocated
  FROM allocation
  INNER JOIN people
    ON people.people_id = allocation.allocation_people_id
  INNER JOIN band
    ON band.band_id = people.people_band
  WHERE allocation.allocation_date BETWEEN '2024-01-01' AND '2024-08-31' -- filter required date
  GROUP BY
    allocation.allocation_people_id,
    allocation.allocation_date,
    allocation.allocation_shift,
    people.people_band,
    people.people_specialty,
    cost_per_shift
  ORDER BY num_wards_allocated DESC
)

SELECT
  people_specialty,
  ROUND(sum(cost_per_shift)/count(allocation_date),2) AS avg_salary_shift_specialty
FROM shift_per_staff
GROUP BY people_specialty
```

Explanation: A CTE "shift_per_staff" is an intermediate table simplifying the final aggregation. Its role is to prepare and organise data per staff member, date and shift before calculating averages by people_specialty. In the temporary table, duplications of wards in the same shift and date are removed and unique by people ID, date, shift, band, and specialty. Moreover, the expression round(band.band_salary/230.0,2) calculates the cost for each shift per member and rounds to 2 decimal places. Because there is no longer duplication, the main query aggregates data using the CTE table. The average staff costs per shift for each specialty are calculated by summing the total cost per shift divided by the total shifts and rounded to 2 decimal places.

Results: 11 rows, 2 columns

month_num	band_type
Cardiology	185.84
Emergency	179.32
General	176.76
Geriatric	159.94
Neurology	187.97
Oncology	173.55
Ophthalmology	185.88
Orthopaedics	165.15
Paediatrics	167.90
Psychiatry	176.40
Respiratory	170.37

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Thank you for reviewing this report!